Hanoi University of Science and Technology - SOICT

Introduction to ICT

(INFORMATION AND COMMUNICATION TECHNOLOGY)



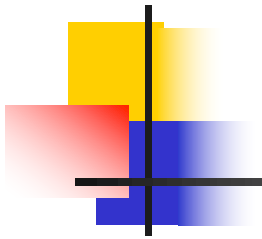
LECTURE 1

Introduction and Overview

Contents

1. Basic Concepts of ICT
2. Information Systems
3. People
4. Hardware
5. Software
6. Data and Information
7. Computer Network and Internet

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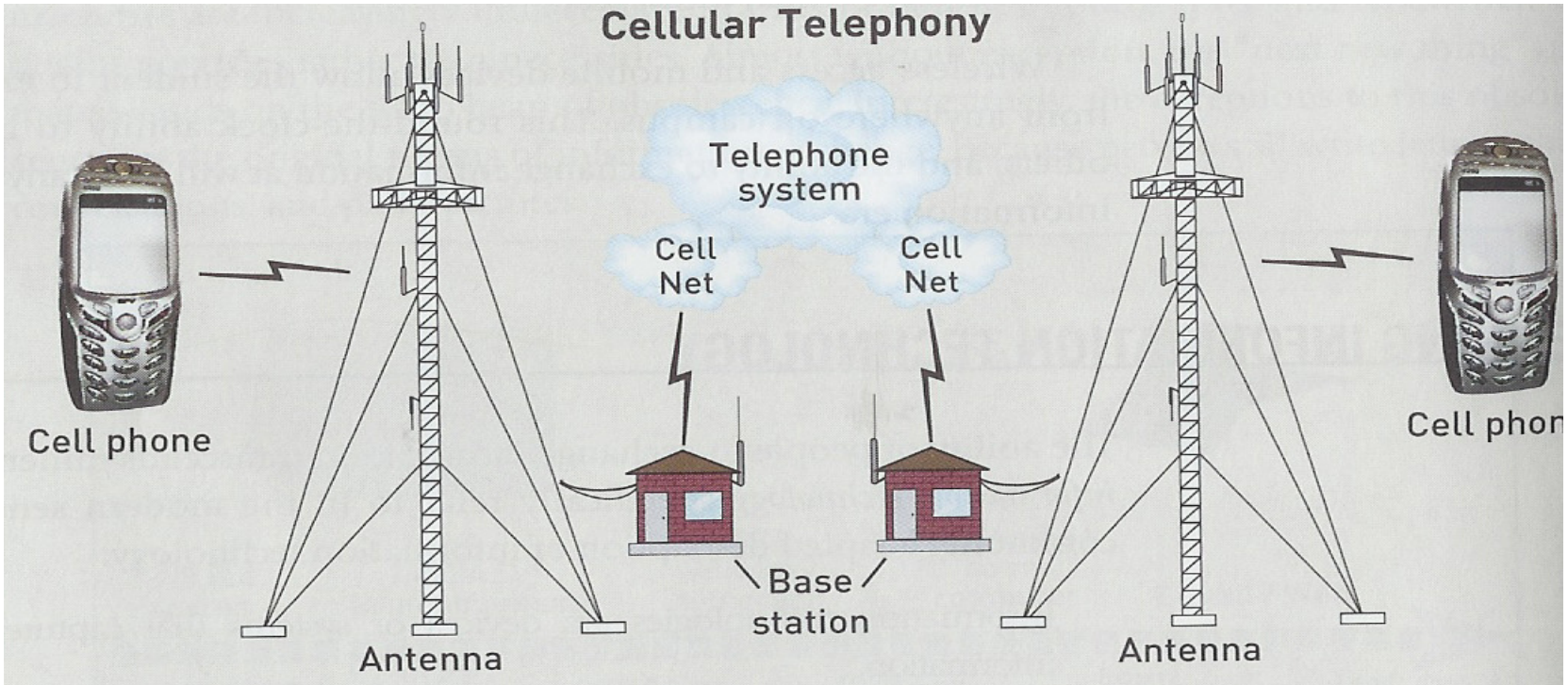
1. Basic Concepts of ICT

- *Informatics (Computer Science):* The science of Computer and information processing on computer.
- *Information Technology* – IT (or so-called *Information Technology and Communication* – ICT recently) is the combination of Informatics and Communication Technology

Defining Information Technology

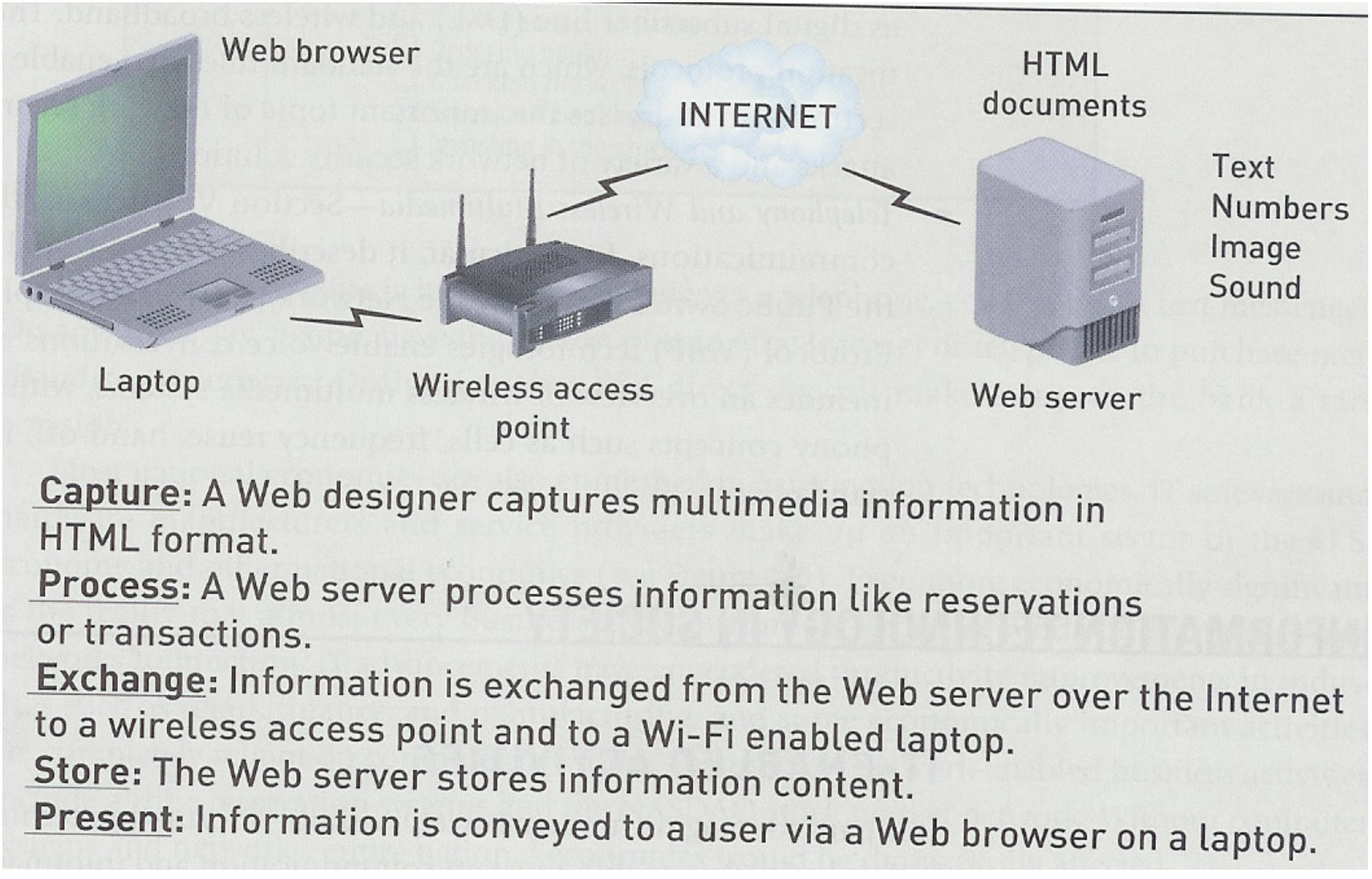
- **Information Technologies (IT)** are systems of hardware and/or software that capture, process, exchange, store and/or present information using electrical, magnetic and/or electromagnetic energy.

IT Example: Cellular Telephony



Capture: A cell phone captures the sound of a human voice and converts it into electrical signals.
Process: Network equipment determines where to route the call.
Exchange: A network routes the call from origination to destination.
Store: A voice mail system stores information for later use.
Present: A cell phone translates information from electrical signals to sound waves the recipient can understand.

IT Example: Wireless Internet



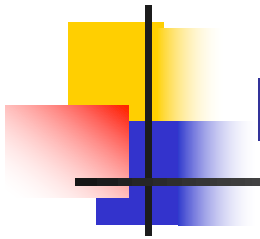
Capture: A Web designer captures multimedia information in HTML format.

Process: A Web server processes information like reservations or transactions.

Exchange: Information is exchanged from the Web server over the Internet to a wireless access point and to a Wi-Fi enabled laptop.

Store: The Web server stores information content.

Present: Information is conveyed to a user via a Web browser on a laptop.



IT in Society (people)

- Personal Communication
  - Conversations (phone, cell)
  - Messaging (E-mail, SMS)
  - Video Coms
- Entertainment
  - Web surfing
  - Downloading video and audio files
  - Interactive gaming
- Day-to-Day living
  - Buying airline tecket
  - Ordering books
  - Electronic banking/ stock market

IT in Society (business)

- Internal Communication
  - Computer network
  - Corporate website
  - Video teleconferencing
  - Messaging (SMS, email)
- Electronic Commerce
  - Call Centres
  - Electronic transactions
  - Online sales
- Business operations
  - Factory operation systems
  - Databases

Examples on IT Careers

- Information Security Jobs
  - Address information security
- IT Analyst
  - Translate business requirement into technical specification
- Network Administration
  - Configure and operate computer network
- Management Consulting
  - Provide consulting for government and organizations
- Database Administration
  - Managing corporate database
- Computer Forensics Expert
  - Extracts computer evidence for detecting/preventing /prosecuting crimes
- IT Sales:
  - people on frontline to sell products
- Software development:
  - develop software solutions

ICT IN DIFFERENT SUBJECTS

A) ICT IN NATURAL SCIENCE

- Spread sheet to tabulate calculate results
- To see visual effect of manipulating variable
- To draw charts
- Composing documents &presentation

B) ICT IN MATHS

- Students can use applets available on internet for performing mathematical problem
- Can use modelling packaging like mathematacia
- Suitable statistical software tools (ex. Excel) to eliminate hours of calculation

C) ICT IN SOCIAL SCIENCE

- For presentation of topic using graphics
- Use spreadsheet for listing dates, events, countries & person involved
- Report making on Social & ethical issues via Internet

2. Information System

Five main parts of an Information System (IS):

- People (end users): the most important factor that contribute to the success of an Information System. Computers are all to make people more productive.
- Procedures: rules or guidelines for people to follow when using software, hardware, and data, always need to be well-documented.
- Software: so-called Computer Programs, consists of the step-by-step instructions that tell the computer how to do its work to convert data into information of vice versa.
- Hardware: The equipment that processes the data to create information, is controlled by software.
- Data: The raw, unprocessed facts, including text, numbers, images, and sounds. Information is made by Data Processing Procedures in Computer.

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Five main parts of an Information System



An additional factor: ***Connectivity*** is becoming more and more important, provide a shared collaboration environment to enable network applications.

3. People



People
Are end users who use computers to make themselves more productive

- The key and most important factor of any Information System.
- Computer was designed to meet people's need, to serve people.
- Requirement:
  - Competent skills with Computer.
  - In-depth know-how of Computer and Information Technology.

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3. People

Computers in Medicine and Education





…human must be the central factor !

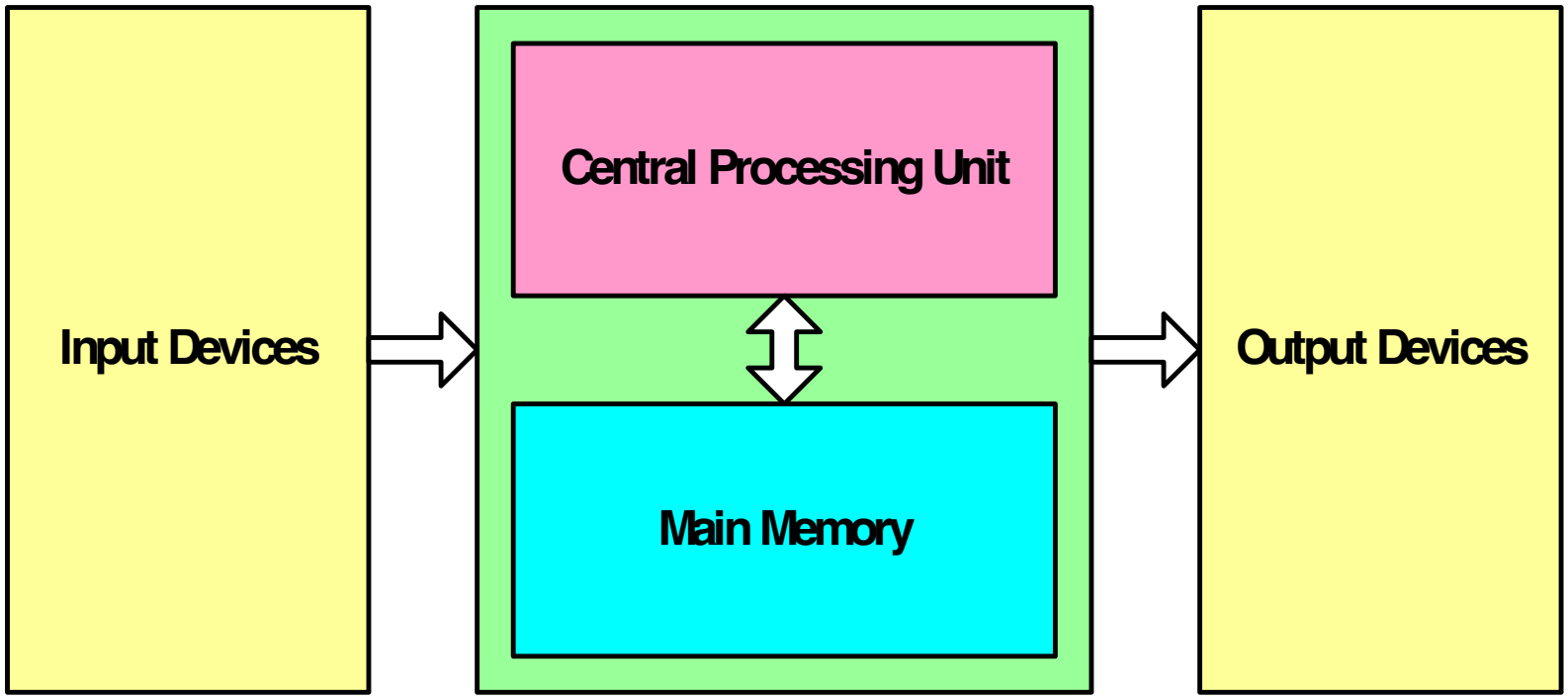
4. Computer Hardware

- Computers are electronic devices that can follow instructions to accept input, process that input, then produce information.
- Computer program is a sequence of instructions stored in memory which is used to guide the computer to manage a specified task

→ Working principle: follow instructions from a stored-program (software).

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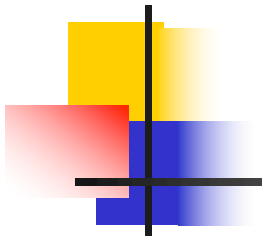
Fundamental model of Computer



Computer classification

- Supercomputer
- Mainframe Computer
- Minicomputer
- Microcomputer

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Supercomputers & Mainframe Computer

- Super fast and expensive
- Suitable for big problem/application which require a huge amount of data processing
- Examples: Deep Blue, Blue Gene
- Typical applications:
  - Earth simulation
  - Weather forecast, earth quakes, tsunamis…
  - Geological simulation to locate oil-fields
  - Nuclear explosion simulation

Supercomputer



The Earth Simulator running simulations of global climate (NEC, Japan 2002)

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Mainframe



IBM eServer zSeries 990 (2003)

Minicomputer

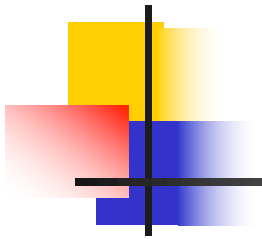


Sun Fire V490 Server



Sun Fire E4900 Server

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Microcomputers

- Desktop Computers
- Notebook Computers, Laptop Computers
- Handheld Computers, Palmtop Computers, PDA - Personal Digital Assistants

Microcomputers









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Microcomputer hardware

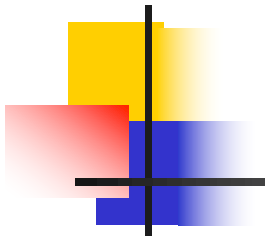
- System Unit: Chassis, Mainboard, Microprocessor, Memory, I/O Cards, Power Supply.
- Input/Output devices: keyboard, mouse, monitor, printer …
- Storage devices: Floppy / Hard disk drives, Optical drives (CD/DVD), Flash drives.
- Communication and Network devices: Modems, NICs…

5. Computer Software

- Software are all computer programs.
- Program is a sequence of instructions to tell the computer how to process data and produce output information following user requirement.
- 2 major kinds:
  - System software
  - Application software



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System software

- **System software** enables the application software to interact with the computer hardware – background software.
  - **Operating systems:** programs that control computer resources, provide users-computer interface, and run applications.
    - Windows XP, MacOS, Linux, ...
  - Utilities: specific tasks like Disk Defragment, Compress/Decompress files
  - Device drivers: control I/O devices

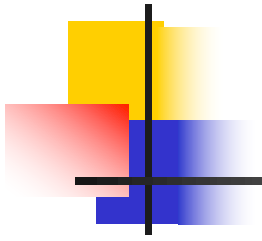
Application software

- **Application software:** end-user software, including *general-purpose and special-purpose applications.*
- There are many types of application software:
  - Office software
  - Management software
  - e-Learning software
  - e-Commerce software
  - e-Library software
  - Graphics & Image Processing software
  - Multimedia software
  - CAD – Computer Aided Design software
  - Communication software
  - Entertainment software
  - Web Pages
  - ...

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6. Data and Information

- Data are raw, unprocessed facts, including text, numbers, images, and sounds. Processed data will become information.
- Information is the data processed with people's requirement.
- All the data must be represent in binary format to be stored and processed by computer.
- Data is stored in computer in files: document, music, video, spreadsheet, database…



Basic data types

- Numbers
- Text
- Sounds
- Images
- …

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Unit of information

- Bit (Binary Digit): basic unit of information in computer, taking a value of either 0 or 1 .
- Byte: a combination of 8 bits, can represent 256 different values (2^8)
- KB (Kilobyte) = 2^{10} bytes = 1024 bytes
- MB (Megabyte) = 2^{10} KB = 2^{20}bytes (10^6)
- GB (Gigabyte) = 2^{10} MB = 2^{30}bytes (10^9)
- TB (Terabyte) = 2^{10} GB = 2^{40}bytes (10^{12})
- PB (Petabyte) = 2^{10} TB = 2^{50}bytes
- EB (Exabyte) = 2^{10} PB = 2^{60}bytes

7. Computer Network and the Internet

- **Connectivity** is the capability to share information between microcomputer over distance or space.
- Computer Network: multiple computers are connected to share data and resources.
- Internet: world-wide computer network.
- World Wide Web (www): a service provides standardized interface to access multimedia resources on the Internet.

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